Hayeon Bang

hayeonbang@kaist.ac.kr
website | github | linkedin

Researcher, Composer

Research interest: Educational Music AI, Human-AI Musical Interaction

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) Feb 2025 - present
Ph.D. student in Graduate School of Culture Technology
Music and Audio Computing (MAC) Lab | Advisor: Prof. Juhan Nam

Korea Advanced Institute of Science and Technology (KAIST) Mar 2022 - Feb 2025
M.S. in Graduate School of Culture Technology
Music and Audio Computing (MAC) Lab | Advisor: Prof. Juhan Nam

Yonsei University Feb 2016 - Feb 2021
Bachelor's Degree in Music Composition

PUBLICATIONS

PitchIn: Natural-Language Vocal Coaching from Unstructured Community Feedback
Hayeon Bang, Hounsu Kim, Joonyong Park, Jung-Jae Yu, Juhan Nam
Under Review, 2026

VocalCoachBench: Benchmarking Audio-Language Models on Expert Feedback for Singing
Hayeon Bang, Hounsu Kim, Wonil Kim, Juhan Nam
Conference on Neural Information Processing Systems (NeurIPS), Evaluation & Dataset Track, 2026

D3PIA: A Discrete Denoising Diffusion Model for Piano Accompaniment Generation
Eunjin Choi, Hounsu Kim, Hayeon Bang, Taegyun Kwon and Juhan Nam
IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), 2026

PianoBind: A Multimodal Joint Embedding Model for Pop-piano Music
Hayeon Bang, Eunjin Choi, Seungheon Doh and Juhan Nam
Proceedings of the International Society of Music Information Retrieval (ISMIR), 2025

Dialogue in Resonance: an Interactive Music Piece for the Piano and Real-time Automatic Transcription
Hayeon Bang, Taegyun Kwon and Juhan Nam
Proceedings of the International Computer Music Conference (ICMC), 2025

PIAST: A Multimodal Piano Dataset with Audio, Symbolic and Text
Hayeon Bang, Eunjin Choi, Megan Finch, Seungheon Doh, Seolhee Lee, Gyeong-hoon Lee and Juhan Nam
Proceedings of the 3rd Workshop on NLP for Music and Audio (NLP4MusA), 2024

PROJECTS

Jarasum Jazz Festival AI Jazz Lounge (2026) **[link]**
Producer for AI-generated Jazz Solo

X-Space: Duo Performance for Piano AI Piano | KAIST X Daejeon Arts Center (2024) **[link]**
Composer, Music Director

EasyJam: AI-based Jam Program for Developmental Disabilities | Selected for the 2024 Gyeonggi-do Integrated Support Program for Artists with Disabilities, with Mirae-therapy center (2024)
Program Developer

Four Seasons 2050 - Daejeon (2023) [link]
Project leader, Composer

AWARDS & HONORS

Seong-bu Kim Creativity Award (2025) | **KAIST**

High Honor Awards: Academic Achievement for highest grade students, Academic Achievement for high grade students (2020, 2021) | **Yonsei University**

Scholarship for Human Resource Development (2021) | Mapo-gu

Scholarship for Exchange Program (2019) | Mirae Asset Park Hyeon Joo Foundation
Representative awardee

SELECTED RESEARCH PROJECTS

AI Agent Technology for Interactive Cultural Creation Based on Artists' Individual Characteristics
Korea Creative Content Agency | 2025–present

AI/Big Data-Based Platform for Process-Oriented Assessment of Middle School Music Performance
Collaboration with Ewha Womans University, Music Education | 2024–2026

Deep Learning Models for Improving Music Tagging Performance
Funded by Kakao Entertainment | 2023–2025

Piano Cover Generation for Pop Music with Text-Based Style Control
Funded by NCSoft | 2023–2025

Music-Making in the Post-AI Era
KAIST | 2023

MUSICAL EXPERIENCE

Singer, Piano Arranger at Sumi Jo Talk Concert, KAIST (2024)

Choral Conductor in Woncheon Church 1st Choir (2021 - present)

Actress, Music Director, President in Yonsei University Musical Club ROTHEMS (2016 - 2020)

Orchestra Conducting Education, Sejong Orchestra Conductor Course (2018)

Musical Theater Education, Chungmu Art Center Musical Theater Academy (2018)

TEACHING EXPERIENCE

Teaching Assistant | GCT731 Topics in Music Technology <Music Informatics>, KAIST (2026)

Teaching Assistant | CTP 431 Fundamentals of Computer Music, KAIST (2026)

Instructor | AI for Music Education, Association of Korean Traditional Music Teachers (2026)

Teaching Assistant | GCT 536 Cognitive Science of Music, KAIST (2025)

Instructor | Sonic Visualizer for Music Composers, Yonsei University (2023)

Instructor | Understanding of Music AI, Dangjin Youth Center (2022)

Teaching Assistant | Sound Technology for Multimedia, KAIST (2022)

Instructor | Music Composition, Music Theory, Ear Training, Piano (2017 - 2024)